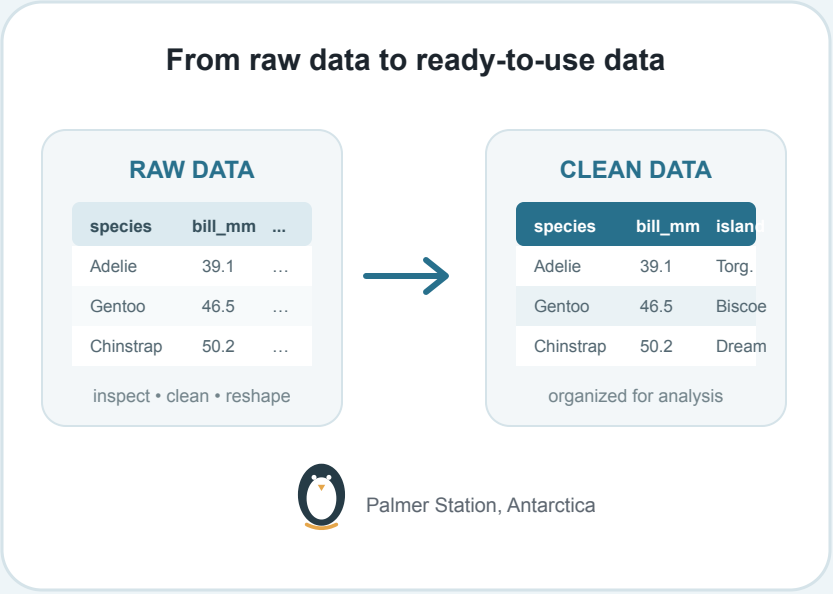
DATA 110 Week 8

Data wrangling 1

Clean, reshape, and prepare data for analysis

Python and pandas, using measurements of Palmer penguins



Prof. Rebin Muhammad

Montgomery College

TODAY

A picture starts as a smaller table

- The file has 344 penguins and 8 columns.
- A plot usually needs some of the rows and some of the columns.
- Today's tools pick rows, pick columns, sort, count, and add a column.

The table comes first, then a dot, then the next step.

PART 1

Look at the table before you draw

One row is one penguin. head shows the first rows, so you can see the column names before you ask for a subset.

The first three penguins

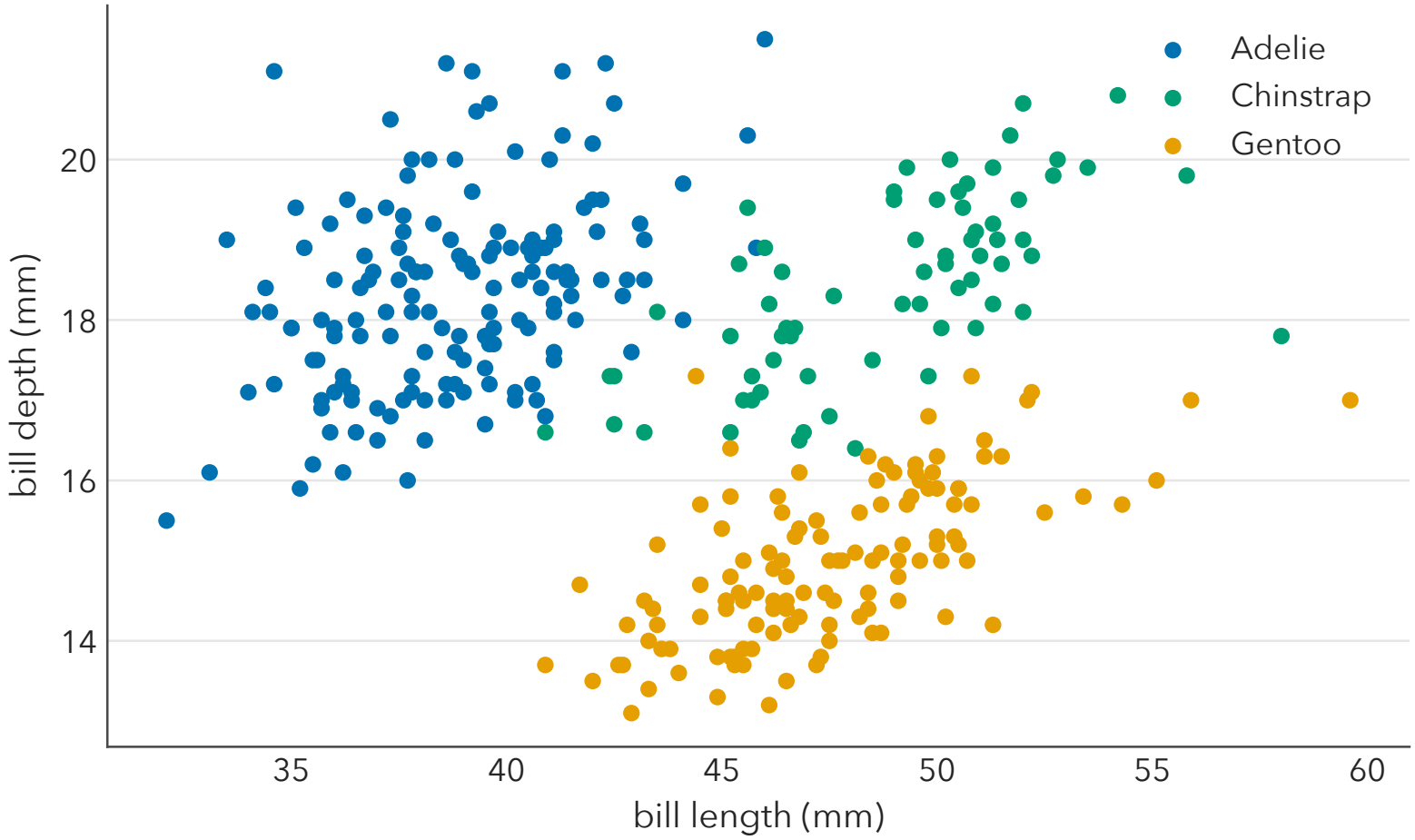
All three are Adelie, from Torgersen, studied in 2007. Bill length and bill depth are in millimeters. Flipper length is in millimeters. Body mass is in grams.

SPECIES	ISLAND	BILL LENGTH	BILL DEPTH	FLIPPER	MASS	SEX
Adelie	Torgersen	39.1	18.7	181	3750	male
Adelie	Torgersen	39.5	17.4	186	3800	female
Adelie	Torgersen	40.3	18.0	195	3250	female

Two penguins have no bill measurements. Eleven have no sex recorded. A blank is not a zero.

Three columns are enough for this picture

Bill length, bill depth, and species. The colors are last week's kind of scale: a name, not a size. Adelie is blue, Chinstrap is green, Gentoo is orange.



PART 2

Pick the rows you mean

`loc` keeps the rows that pass a test. The original table stays complete.

The test writes True or False on every row

These four species names are real. They are not the first four rows of the file. `==` asks whether two values are equal. `=` stores a name. The test uses `==`.

SPECIES	SPECIES == "GENTOO"
Adelie	False
Gentoo	True
Chinstrap	False
Gentoo	True

The spelling has to match the table. `"gentoo"` matches nothing, because the table says `"Gentoo"`.

QUESTION

How many of these four rows stay?

The test column is True, False, False, True. `loc` keeps a row only when the test is True.

Two rows stay

The two Gentoo rows. Adelie and Chinstrap are False, so they are left out of the new table. The same test on the full file keeps 124 rows.

Gentoo is 124 of the 344 birds

Every Gentoo in this file was measured on Biscoe. These are the first four.

SPECIES	ISLAND	BILL LENGTH	SEX
Gentoo	Biscoe	46.1	female
Gentoo	Biscoe	50.0	male
Gentoo	Biscoe	48.7	female
Gentoo	Biscoe	50.0	male

QUESTION

How many rows does penguins still have?

The Gentoo result has 124 rows. What is left in the table named `penguins`?

Still 344

`loc` builds a new table. It does not delete rows from `penguins`. Give the new table its own name if you want to use it again.

```
gentoo = penguins.loc[penguins["species"] == "Gentoo"]
```

A missing bill is not greater than 50

52 penguins have a bill longer than 50 mm. Two more have no bill length recorded. A comparison with a blank is false, so those two rows stay out. They are not short bills. They are missing bills.

```
long_bills = penguins.loc[penguins["bill_length_mm"] > 50]
```

Two tests, both required

Chinstrap and Dream together. In this file every Chinstrap lives on Dream, so the result is all 68 Chinstraps. The & means both tests have to be true. Each test needs its own parentheses.

```
chinstrap = penguins.loc[
    (penguins["species"] == "Chinstrap")
    & (penguins["island"] == "Dream")
]
```

PART 3

Pick columns, then put the rows in order

A list of names keeps those columns. `sort_values` reorders the rows.

Three names, three columns

The new table has species, bill length, and flipper length. Bill depth, mass, sex, and year are still in `penguins`. They are absent from `chosen`.

```
chosen = penguins[["species", "bill_length_mm", "flipper_length_mm"]]
```

One name uses one pair of brackets: `penguins["species"]` is the column of 344 names. Several names are a list, and that list goes inside the brackets, so you see two pairs.

The shortest bill is an Adelie, 32.1 mm

`sort_values` orders the rows from small to large. The two missing bills go to the end. They are not the shortest bills.

```
by_bill = penguins.sort_values("bill_length_mm")
```

`ascending=False` reverses the order. The longest bill is 59.6 mm, a Gentoo.

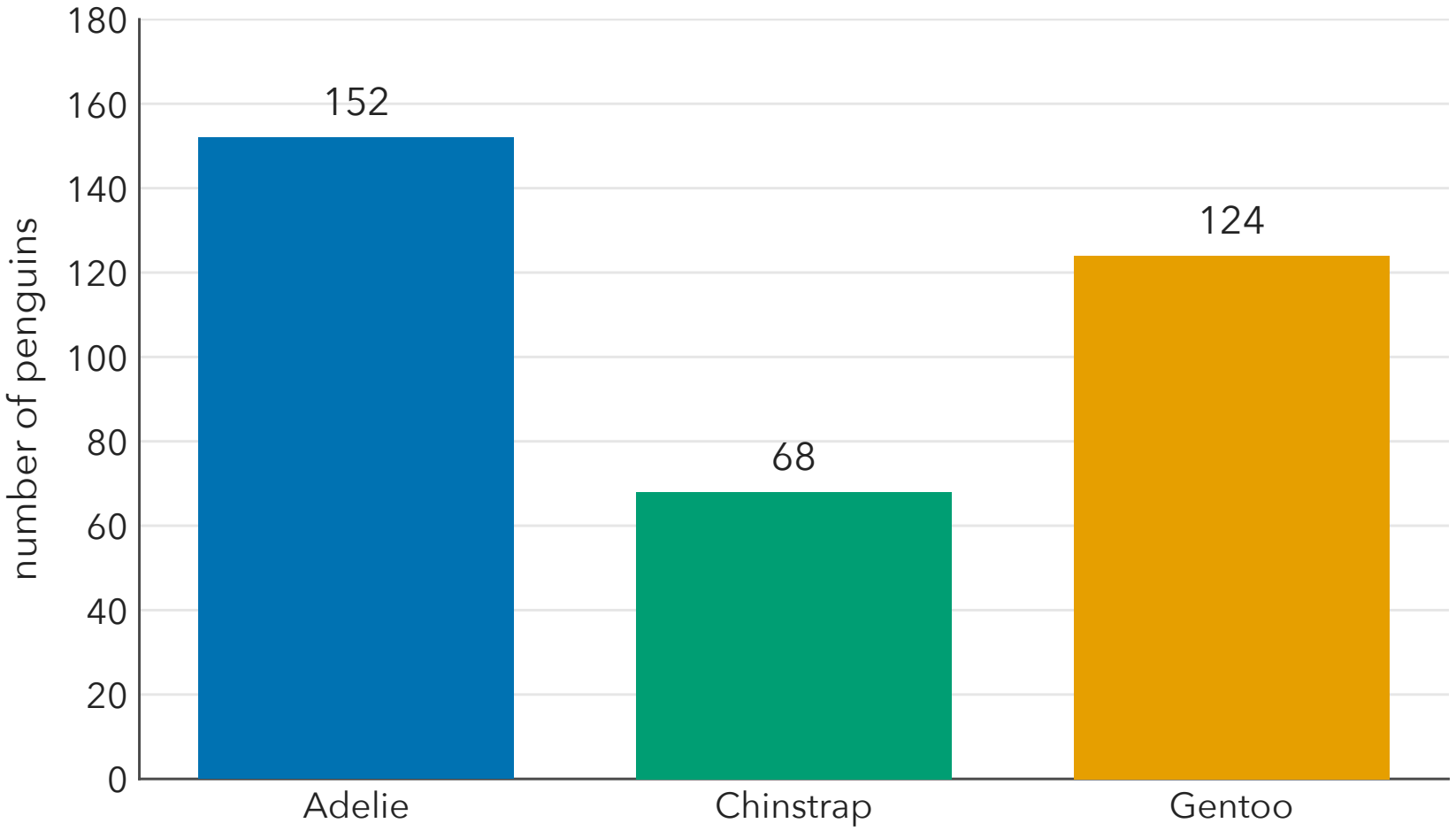
PART 4

Count the rows that share a value

`value_counts` counts one column. `groupby` counts a combination of columns.

Adelie is 152 birds, not 152 islands

`value_counts` counts rows. It puts the most common value first. Gentoo is 124. Chinstrap is 68. The three numbers add to 344.



QUESTION

Where do the Chinstrap penguins live?

Adelie appears on Biscoe, Dream, and Torgersen. Gentoo appears on Biscoe. Which island has the Chinstrap rows?

Every Chinstrap row is Dream

SPECIES	BISCOE	DREAM	TORGERSEN
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

`groupby` splits the table by species and island. `size` counts the rows in each piece. A zero here means that combination does not occur. It is not a missing measurement.

Each line uses the table from the line before

Keep the Adelie rows, keep two columns, then count. Chinstrap cannot appear, because the first line already left those rows behind.

```
adelie = penguins.loc[penguins["species"] == "Adelie"]
chosen = adelie[["island", "sex"]]
chosen.groupby(["island", "sex"], dropna=False).size()
```

`dropna=False` keeps a row whose sex is blank. Dream has 27 female and 28 male Adelie penguins, and one Adelie on Dream with no sex recorded.

PART 5

Add a column the table does not have yet

The new column is computed from columns that are already there. The old columns stay.

Flipper length in centimeters

Divide the millimeters by 10. A 181 mm flipper is 18.1 cm. The millimeters column is still in the table. `copy` makes a table of our own, so the new column is not written back into `penguins`.

```
flippers = penguins[["species", "flipper_length_mm"]].copy()
flippers["flipper_length_cm"] = flippers["flipper_length_mm"] / 10
```

Bill length divided by bill depth

The ratio is one number for the shape of the bill. A larger ratio is a longer bill relative to its depth. The two missing bills make two missing ratios. Dividing by a blank does not invent a shape.

```
bills = penguins[["species", "bill_length_mm", "bill_depth_mm"]].copy()
bills["bill_ratio"] = bills["bill_length_mm"] / bills["bill_depth_mm"]
```

KEEP THESE

What the steps were for

- **loc.** A test writes True or False on every row, and loc keeps the True rows. == asks the question. The original table still has 344 penguins.
- **Brackets.** One pair takes one column. Two pairs, with a list of names inside, take several columns. The columns you leave out stay in the original table.
- **sort_values.** Orders the rows. Missing measurements go to the end, and they are not the smallest values.
- **value_counts and groupby.** Count rows. A zero combination is a combination that does not occur.
- **A new column.** Compute it on a copy. The columns it came from stay as they were.

Further reading

The same steps, in the pandas names

- Allison Horst, Palmer penguins
- pandas, indexing and selecting data
- pandas, value_counts and groupby